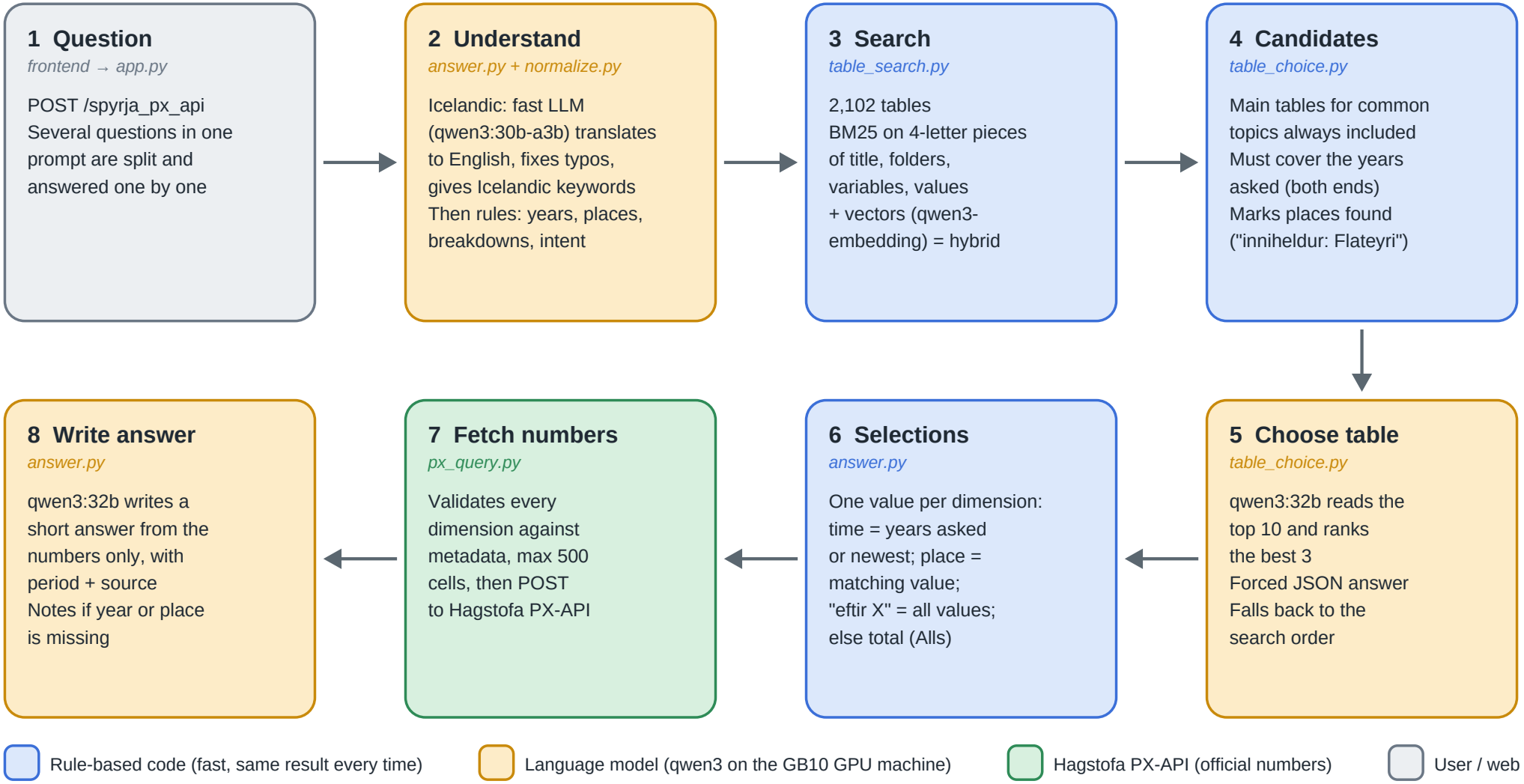
From question to official numbers

Our Track 2 prototype answers questions about Icelandic statistics in Icelandic or English. The numbers always come straight from Hagstofa's PX-API; the language model only **reads the question**, **chooses the table** and **phrases the answer**.



Example: “Hvað áttu margir heima á Flateyri 1990?”

What each step produces for one real question (about 10 seconds in total).

Step	What happens	Result
2 Understand	The fast model translates the question (fixing typos and inflected place names) and suggests Icelandic title keywords; rules then pull out the year and place.	“How many people lived in Flateyri in 1990?” · years: [1990] · keywords: flateyri, mannfjöldi
3 Search	Hybrid search over 2,102 tables. Flateyri is found among the value labels of locality tables.	Population tables by locality (byggðakjarnar), old and new
4 Candidates	Tables that start after 1990 (most locality tables start in 1998) are dropped.	MAN00104 … 1889–1990, marked “inniheldur: Flateyri, Flateyrarhr.”
5 Choose table	qwen3:32b reads the 10 candidates and ranks the best three.	#1 MAN00104 – Mannfjöldi í einstökum byggðakjörnum … 1889–1990
6 Selections	Whole-word match on the place (not Flatey á Breiðafirði); year from the question.	Byggðarkjarni: Flateyri, Flateyrarhr. · Ár: 1990
7 Fetch numbers	Query validated against the table's metadata, then sent to the PX-API.	Flateyri, 1990 → 396
8 Write answer	The model writes one sentence from the number, with the period and the source.	One sentence with 396, the year 1990 and Flateyri, then “Heimild:” with the table name and link

Why it is built this way

• **Numbers never come from the language model.** They are fetched from the PX-API and passed to it; the answer must use only those.

• **Every answer shows its source**: table name, link, the chosen dimensions and the raw data rows (expandable in the web page).

• **Honest when data is missing**: if the year or the place is not in the table, the answer says so instead of guessing (e.g. a question about 2030 gets the newest year, 2026, with a note).

• **Rules plus LLM where it helps**: rules are fast and predictable; the LLM handles the ambiguous parts (understanding everyday or misspelled questions, choosing between similar tables, phrasing).

• **Keeps working if the LLM fails**: table choice falls back to the search order, and the answer falls back to showing the data.

Results

How often the right table is chosen first, on the 83 questions in AI_test/spurningar.txt (analytical questions written by a team member) and our 24 shorter questions.

Method	83 questions	24 questions	Time per choice
Search alone (hybrid BM25 + vectors)	35	12	0.3 s
+ qwen3:4b choosing (laptop CPU), 40-question sample	24 of 40 (search alone: 23)		≈ 25 s
+ qwen3:30b-a3b choosing (GPU)	39	17	0.7 s
+ qwen3:32b choosing (GPU) – used	44	17	2–3 s
Right table anywhere in the top 10	55	21	–

Tested end to end (through the web API)

Question	Table	Answer
Hvað búa margir í Kópavogi?	MAN02005	40,286 (1 Jan 2026)
Hvað bjuggu margir á Íslandi árið 2024?	MAN00000	383,726 (1 Jan 2024)
Hver var verðbólgan á Íslandi árið 2024?	VIS01000	4.8 % (12-month change, Dec 2024)
Hversu margir fluttu til Íslands árið 2024?	MAN01400	19,789 immigrants, 15,745 emigrants
Hvað fæddust mörg börn 2023?	MAN05203	4,315
Hvað áttu margir heima á Flateyri 1990?	MAN00104	396
How many people live in Reykjavík?	MAN03270	139,804 (1 Jan 2026)
Hvað búa margir á Íslandi árið 2030?	MAN00000	2030 not available; newest is 2026
asdkj qwpe	–	“Ég skildi ekki spurninguna …”

Known limitations and next step

• **Understanding and selections are rule-based.** New phrasings need new rules, and if a question's words don't look like the table's value labels, the total is used. **Next step:** let the LLM pick a value for each dimension from the table's metadata, validated by px_query.

• Monthly tables asked about a whole year use the last month: right for stocks (population, inflation), not for flows (passengers in a year would need a sum).

• Answers take 5–25 seconds; English questions are slowest (extra translation step). The model's Icelandic grammar is not perfect.

• Hagur (hagstofa-ai:latest) was not available to us; the model is a setting (MODEL=…), so it can be swapped in.